

V6: RAP Application Patterns

Time 11:30–12:15 Start 11:30 Track 2 Duration 45 min Language en

ABAP

Clean Code

experienced

expert

Session

Session description:

Clean Core compliant Custom Fiori Apps combining advantages of RAP, Application Jobs and Logs

Application Jobs allows the execution on mass data specified via select options including a nice integration of application logs. On the one hand application jobs are very technical and not well-accepted by end-users, on the other hand Fiori apps (e.g. based on RAP) do not offer mass-data processing.

With our application blueprint we filled the gap and combined furthermore the benefits of both approaches: RAP-based Fiori Apps and Application Jobs and Logs.

Speaker



Marcel Hermanns

[Speaker Profil →](#)



Brandeis
Consulting

Custom Development Project

S/4HANA

RISE

Greenfield

Automated Business Processes

&

Mass Change Enablement

Custom Development Project

1. Automated Business Processes

- **Challenge:** Broken process chain
- **Approach:** Error-tolerant and self-healing design via retry mechanism with advanced error handling

2. Mass Change Enablement

- **Challenge:** End-user friendly app with visualization of progress and result and detailed log
- **Approach:** Combination of RAP and Fiori Elements with Application Jobs and Logs

User Story

Rate Adjustment Run

User Story

- 1 As a contract manager I want to trigger an **adjustment run for the service contract items** that have a specific material and assigned equipment category with a discount in % in an **end-user friendly app**.
- 2 After triggering the execution I want to see the **progress** and later a **summary** of the results,
- 3 with the option to navigate to a **detailed application log** with search and sort capabilities.
- 4 I also want to easily **find the adjustment run** results with filtering on user, sorting on date and searching for parameter values.
- 5 It should also be possible to **plan the adjustment run** by expert users.

1 Initiate Run

2 Run results

3 Run log

4 Run list

5 Run Job

Mass Change Enablement

Service Contract

Sold-to-party

Equipment Category

Service

Discount (%)

☐ Run in background

ABAP Report

1 Job 2 Date / time 3 Parameters

Service Contract

Sold-to-party

Equipment Category

Service

Discount (%)

Application Job

Service Contract

Sold-to-party

☐							
☐							
☐							
☐							

Fiori App

Look & Feel	-	SAP GUI Transaction	-	Job specific, very technical	+	End-User friendly Fiori
Mass changes	+	Select Options + Parameters	+	Select Options + Parameters	-	No Select Options
Data Set	-	No visualization	-	No visualization	+	List report shows list according to filter values
Visualization					+	Can be freely defined with object-based navigation
Navigation to Results	+	Can be freely defined	-	After job creation navigation to list	+	Fiori apps are desired in ABAP Cloud
ABAP Cloud Clean Core	-	No SAP GUI in ABAP Cloud	+	Application Jobs are released for ABAP Cloud		

Design Idea: Combine benefits

1 End-user friendly Initiation

Service Contract

Sold-to-party

☐							
☐							
☐							
☐							

Equipment Category

Service

Discount (%)

☐ Run in background

2 Progress and Summary

Progress Indicator
#ERRORs
#SUCCESS

Message Severity
One Message

3 Detailed Log

Search

- + Service Contract 01 – 001: adjusted
- + Service Contract 02 – 002: adjusted
- Service Contract 03: locked by another user
- Service Contract 04: Billing plan.....

A Apply ABAP Cloud technologies

RAP & Fiori Elements

Application Jobs, ABAP Parallel, Application Logs

B Add missing building blocks

Adjust via Select Options

Persist progress and summary

Search for Parameter

C Generalize as blueprint

for 9 use cases

in 1 of 10 projects

D Outlook: ? Generalize as “RAP Application Pattern – RAP-AP1“ ?

Best practice built on real customer projects

Available in GitHub – OpenSource for S/4 2023, S/4 2025, BTP ABAP and S/4 Public

Demo!!

End User Perspective

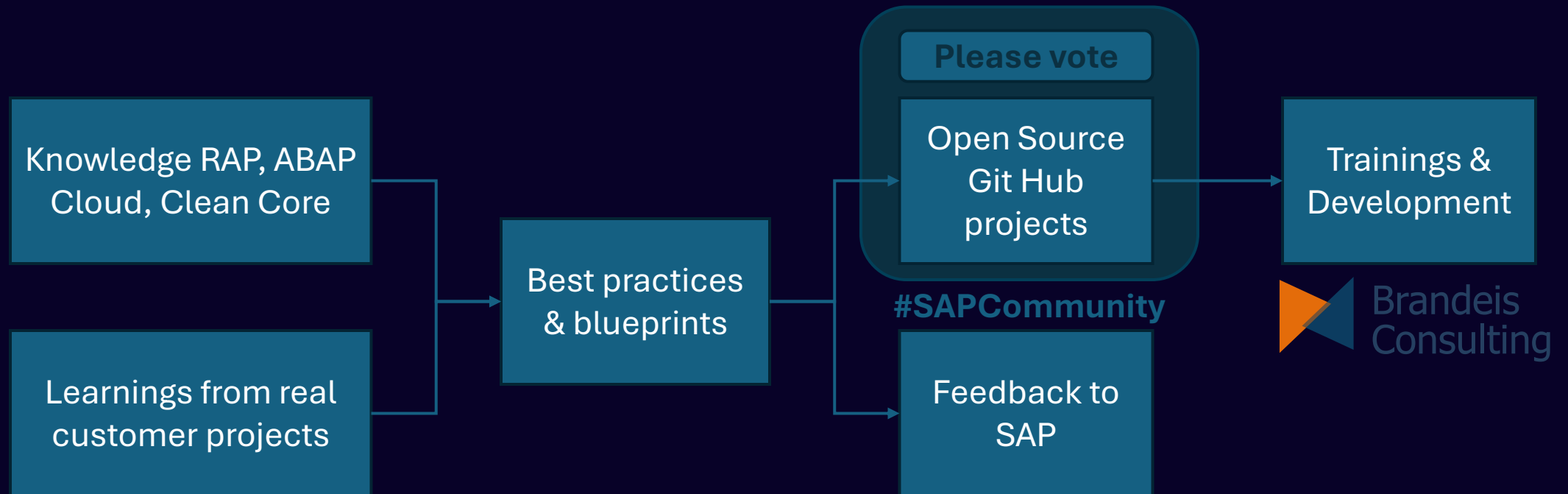
- Application Job
- Initiate App, Run App, Log
- Run App (parameter search)
- Fun fact: Multiple runs

Developer Perspective

- App navigation
 - ADT: CDS annotations
 - BAS: Fiori Elements extensions
- Select Options
 - BAS: Fiori Elements extensions
 - ADT: CDS Modeling, Run BO, Application Job
- Progress
 - Application Job
 - Run BO, Event-based Side-effects

Idea: RAP Application Patterns

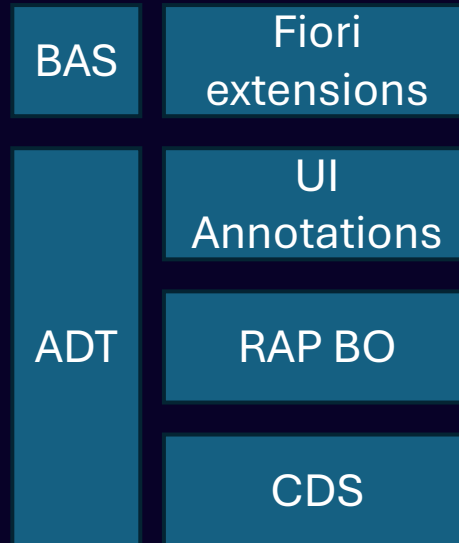
Blueprints for re-occurring custom application patterns



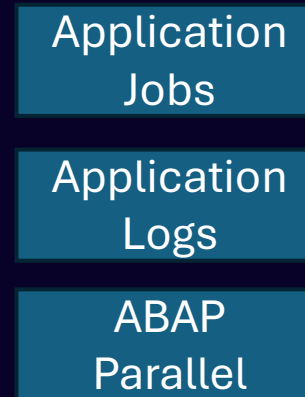
RAP Application Pattern – RAP-AP1 Fiori Mass Changes

#SAPCommunity

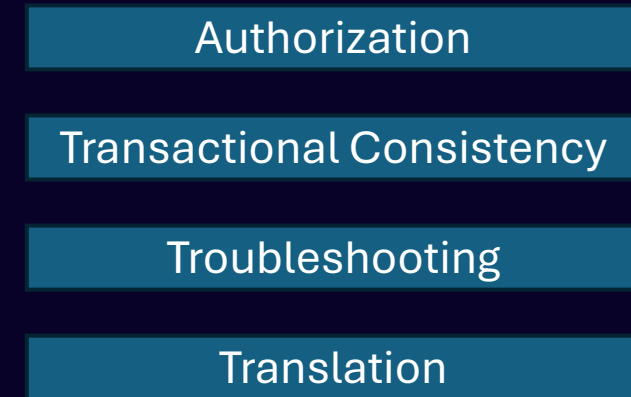
Fullstack Development –
RAP and Fiori Elements



Reuse
Components



Non-functionals



Clean Core

RAP Application Pattern – RAP-AP1 – Mass Changes

S/4HANA Cloud Private Edition / On-Premise 2023 & 2025

SAP BTP ABAP Environment

S/4HANA Cloud Public Edition

RAP Application Pattern

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RAP-AP1 Fiori Mass Changes (Fiori, RAP, Application Jobs, Application Logs, ABAP Parallel)
RAP-AP2 Reliable event handling (Business Events, Application Jobs, Mails, Fiori Notifications)
RAP-AP3 Reliable BAPI call chaining (RAP, BAPI, transactional consistency)

Let's make this happen!

I need 100 persons around the globe to vote.

Please vote now!

Deadline: 1st August



RAP Application Patterns

Thanks for your valuable time!



Marcel Hermanns

